Lesson plan · for the teacher

Exploring a dataset

Data science · Level 2 · Lesson 4

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Follow a five-question routine.
- Spot gaps and oddities.
- Write findings with evidence.

Activity: Five questions, three findings

1. Warm-up (5 min): Show a big table of attendance by class and month. Ask: what is the first thing you would look at?
2. Teach (8 min): Walk through the five questions on a small table, drawing a histogram and a scatter plot.
3. Explore (15 min): Each group works through the five questions with its own table, noting oddities.
4. Write (7 min): Groups write three findings with numbers and sources, then share one.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Which question found the most surprises?
- Why check oddities before comparing groups?
- What is your next question?

Exit ticket

- Write one finding from today with a number and a source.

Try it in class tomorrow
Give groups a small school table and the five questions on a card. Each group writes three findings with evidence and one next question.

Worksheet · for pupils

Exploring a dataset

Data science · Level 2 · Lesson 4

Name _____ Class _____ Date _____

1 **Which is the better finding?**

◯ Attendance is bad ◯ Attendance falls to a median of 80% in July

◯ Attendance changes during the year

2 **Put the five exploring questions in order.**

☐ Groups and relationships

☐ Gaps and oddities

☐ Centre and spread

☐ What shape?

☐ What is in it?

3 **Write summary(xs) so it returns the smallest value, the median and the largest value. Assume xs has an odd number of values.**

```
def summary(xs):
    return 0, 0, 0

print(summary([9, 5, 52, 8, 6]))
```

Write your program:

4 **A column of journey times is skewed right. Which summary should you report?**

◯ The median and the range ◯ Only the mean ◯ The largest value

5 **Which are good findings? Choose all that apply.**

☐ Median attendance is 80% in July and 95% in March (register, 2025). ☐ Our pupils are lazy.

☐ Three classes have missing marks for test 2. ☐ Data is important.

Which school table would you explore first with the five questions? What do you expect to find?